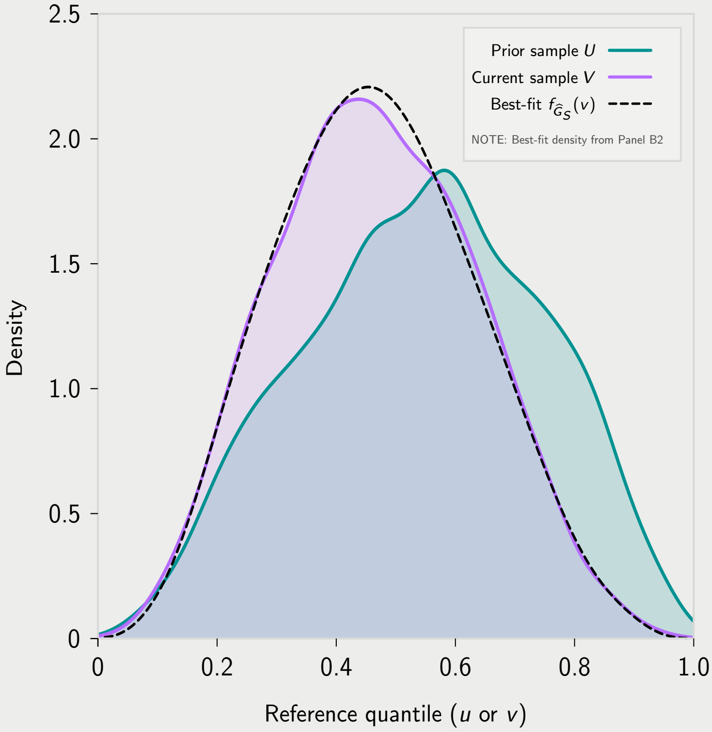
A. Convert scores to reference quantiles

Observed, non-linked marginals: U (prior) and V (current)



We observe two independent cross-sectional, non-longitudinal samples of the same subgroup S (e.g., a TIMSS country or a NAEP state): prior scores X at time t_1 and comparison scores Y at time t_2. The design does not require that t_2 represent a later measurement of the same cohort: (t_1, t_2) may index different grades within the same year (e.g., Grade 4 vs. Grade 8), or the same grade measured in different years (often interpreted as $t_2 = t_1 + 4$ under a same-cohort, four-year-later convention).

We map scores to reference-quantiles (pseudo-observations) using fixed reference marginal CDFs:

1. Prior reference quantiles $U = F_X^{\text{ref}}(x_i)$ and
2. Current reference quantiles $V = F_Y^{\text{ref}}(y_j)$.

Crucially, there is *no student-level linkage* between these samples—we cannot match any u_i to its corresponding v_j.

Panel A shows the two *univariate* marginals plotted on the same reference-quantile axis ("u or v" in $[0, 1]$). This is the information available in TIMSS/NAEP-style designs when cohorts are sampled independently (with survey weights, when applicable).[1]

After the reference-quantile transform, U and V summarize the observable evidence. All subsequent inference proceeds from these marginals plus an externally supplied copula dependence structure.

[1] Panel A also shows the density, $f_{\hat{G}_S}(v)$, the density induced from the baseline copula and fitted growth regime from panels B1 and B2, superimposed on the observed density, $f_{V,S}^{\text{obs}}(v)$.